Viren Variya

Question 1.1

Question 1. (5 Marks = 2.5 * 2)

a. (2.5)

0.5 marks for the correct order of evaluation, and

2 marks (**0.25 × 8** for each truth value combination) for the final answer.

b. (2.5)

0.5 marks for the correct order of evaluation, and

2 marks (**0.25 × 8** for each truth value combination) for the final answer.

Question 2. (5 Marks)

Statement: Prove that if two propositions are equivalent, then they have the same DNFs.

Mark Distribution

- **1 mark**: Choice of method (truth table OR algebraic)
- **2 marks**: Correct explanation of equivalence
- **2 marks:** Argument why DNF must be the same

Saarim Khan

Question 1.2

Question 1. (6 Marks = 1.5 * 4)

(a) (1.5 marks)

(b) (1.5 marks)

(c) (1.5 marks)

(d) (1.5 marks)

Marking Criteria:

- **1.5 marks**: Correct answer with correct justification shown
- **1 mark:** Correct answer with incomplete justification
- **0.5 mark**: Correct answer with incorrect or without justification
- **0 marks:** Incorrect answer with or without justification

Question 2 (4 marks)

Marking Criteria:

- **4 marks:** Complete analysis showing both parts of predicate, correct reasoning about quantifiers
- **3 marks:** Correct final answer with partial analysis of one part.
- **2 marks:** Correct final answer with incomplete analysis
- **1 mark:** Attempt at analysis with some correct elements
- **0 mark:** incorrect answer with or without analysis.

Prashant Shrotriya

Question 1.3 Rubric:

1. This can be solved either by constructing a truth table or by applying the laws of propositional logic.

Method 1: By the laws of propositional logic

DNF Construction(4 Marks) and CNF Construction(4 Marks)

● 1M for each correct step

Method 2: By Truth table

- 4 Marks for the correct truth table and 0.5 marks for each 8 expressions
- Partial marks for an incomplete Truth table.

2.

- 0M for stating “No”.
- 2M for stating “Yes” and a Correct explanation.

Kanu Priya

Ques-1.4

Ques-1.

Rubric:

4 marks if the complete solution is correct

3 marks, solution is correct but rules are not mentioned like De Morgan’s law.

-1 if all steps are correct but final ans is wrong.

Ques-2.

2 marks – Correct answer: No.

1 mark – Correct justification: quantifiers can be expressed as finite conjunctions (for $\forall$) or finite disjunctions (for $\exists$).

1 mark – Clarity of reasoning (possibly with an example, e.g. if domain = {a,b}, then $\forall x\, P(x) \equiv P(a) \land P(b)$).

Ques-3.

2 marks: Topic is mentioned (any valid topic: propositional logic, proof techniques, set theory connections, etc.).